Yujia Liu - CV/Resume

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EDUCATION

Carnegie Mellon University, Pittsburgh, USA — Aug 2025 – Now
Ph.D. in Human Computer Interaction
- Advisor: *Prof. Alexandra Ion*

Tsinghua University, Beijing, China — Aug 2022 – July 2025
M.A. in Information Art and Design
- Advisor: *Prof. Yingqing Xu* and *Prof. Chun Yu*

Tsinghua University, Beijing, China — Aug 2017 – July 2022
BEng in Automation Engineering & BA in Industrial Design (Double Major)
- Advisor: *Prof. Hong Wang*, *Prof. Yingqing Xu*, and *Prof. Lei Zhang*

RESEARCH INTERESTS

Areas	Fabrication, Human-AI Interaction, Education
Methods	Metamaterials, AI Systems, LLMs, Engineering, 3D Printing, Grounded Theory

PUBLICATIONS

[1] BrickSmart: Leveraging Generative AI to Support Children's Spatial Language Learning in Family Block Play.
Yujia Liu*, Siyu Zha*, Yuewen Zhang, Yanjin Wang, Yangming Zhang, Qi Xin, Lunyiu Nie, Chao Zhang, Yingqing Xu.
(ACM CHI'25). In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [Link]

[2] Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process.
Siyu Zha*, **Yujia Liu***, Chengbo Zheng, Jiaqi Xu, Fuze Yu, Jiangtao Gong, Yingqing Xu.
(ACM CHI'25). In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [Link]

[3] Xstrings: 3D printing cable-driven mechanism for actuation, deformation, and manipulation.
Jiaji Li, Shuyue Feng, Maxine Alexandra Perroni-Scharf, **Yujia Liu**, Emily Guan, Guanyun Wang, Stefanie Mueller.
(ACM CHI'25). In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [Link]

[4] 3D-Mirrorcle: Bridging the Virtual and Real through Depth Alignment in Smart Mirror Systems.
Yujia Liu, Qi Xin, Chenzhuo Xiang, Yu Zhang, Lunyiu Nie, Xuhai Xu, Yingqing Xu.
(arXiv). [Link]

[5] MindShift: Leveraging Large Language Models for Mental-States-Based Problematic Smartphone Use Intervention.
Ruolan Wu, Chun Yu, Xiaole Pan, **Yujia Liu**, Ningning Zhang, Yue Fu, Yuhan Wang, Zhi Zheng, Li Chen, Qiaolei Jiang, Xuhai Xu, Yuanchun Shi.
(ACM CHI'24) In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. [Link]

[6] KeyFlow: Acoustic Motion Sensing for Cursor Control on Any Keyboard.
Yujia Liu*, Qihang Shan*, Zhihao Yao, Qiuyu Lu.
(ACM UIST'24 Poster) In *Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. [Link]

[7] FlexEOP: Flexible Shape-changing Actuator using Embedded Electroosmotic Pumps.
Tianyu Yu, Yang Liu, **Yujia Liu**, Qiuyu Lu, Teng Han, Haipeng Mi.
(ACM UIST'24 Demo) In *Adjunct Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. [Link]

[8] More Than Shapes: Exploring the Tactile Parameters of Art Appreciation for the Visually Impaired.
MingYu Cui, Chao Yuan, **Yujia Liu**, Yingying Zheng.
(ACM UbiComp'24 Workshop) In *Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*. [Link]

PROJECT EXPERIENCE

[P.1] [Lead] 3D Block Play Instruction Generation and Conducting in real family learning. — 10/2023 - Present
- Led a project to generate personalized block play instructions for creativity and eco-friendly reuse of bricks.
- Designed an AI Agent for children's spatial language learning in family block play, using LLMs and Gen-AI.
- Conducted a lab experiment with 24 parent-child pairs (children aged 6-8), demonstrating the system's effectiveness in enhancing spatial language skills.
- First author of this paper, accepted by ACM CHI'25.

[P.2] [Lead] LLM Agent for Mentoring Students in the Creative ProblemSolving Process. 05/2024 - Present

- Co-led the development of Mentigo, an interactive agent using LLMs to mentor middle school students through the creative problem-solving (CPS) process, with dataset of real classroom interactions and an agentic workflow.
- Tested effectiveness through a comparative experiment with 12 students and feedback from 5 expert teachers, showing significant improvements in student engagement and creative outcomes.
- Co-first author of this paper, accepted by ACM CHI'25.

[P.3] [Lead] Enhancing AR in Smart Mirrors with Depth-Aligned 3D Visualization 08/2022 - Present

- Led the development of 3DMirrorcle, a system addressing depth mismatch in AR smart mirrors using lenticular grating for a 3D display and algorithms to align AR content with users' reflections.
- Conducted user studies across various tasks and scenarios, demonstrating superior performance in user experience compared to existing solutions.
- First author of the work.

[P.4] [Main Contributor] 3D Printing Cable-Driven Mechanism 06/2024 - Present

- We developed Xstrings, a method for 3D printing cable-driven mechanisms in a single process, enabling four types of interactions: bend, twist, coil, and compress, activated by applying force to the cables.
- Investigated the impact of various printing parameters on maximum tensile strain and the repeatability of interactions without cable failure.
- This work has been accepted by ACM CHI'25. My work included mathematical derivation, test printing parameters, engineering the prototype, and writing parts of the paper.

[P.5] [Lead] Using Acoustic Motion Detection for Cursor Control on Keyboard 05/2024 - 07/2024

- Led the development of KeyFlow, a system that integrates mouse functionality into keyboards using machine learning, enabling users to control the cursor by gliding their fingers across the surface without pressing keys.
- Our research shows that KeyFlow reduces hand movement by 78.3%, significantly enhancing typing efficiency.
- First author of work published at ACM UIST'24 Poster.

[P.6] [Main Contributor] Flexible Shape-changing Actuator using Embedded Electroosmotic Pumps 04/2024 - 07/2024

- We developed FlexEOP, a method for creating fully flexible electroosmotic pumps, enabling adaptable, self-contained shape-changing actuators.
- FlexEOP's versatility is demonstrated in applications such as flexible displays, panels, curved surfaces, and soft robotic fibers.
- This work has been published on ACM UIST'24 Demo. My work contributions include experimental design and testing, modeling and rendering, and writing parts of the paper.

[P.7] [Main Contributor] An Aesthetic Education Workshop for the Visually Impaired 04/2024 - 07/2024

- We enhanced art education for the visually impaired by focusing on Impressionist paintings through workshops.
- Experts translated key painting elements (layout, content, color, lighting, brushwork) into tactile forms, using clay modeling to help participants experience, analyze, and create art, enriching their engagement.
- This work has been published on ACM UbiComp'24 Workshop. My work was method development, paper writing.

[P.8] [Main Contributor] Leveraging LLMs for Context-Aware Interventions in Digital Wellbeing 11/2022 - 09/2023

- We developed MindShift, a mobile app that uses LLMs to create dynamic, personalized content aimed at reducing problematic smartphone use, adapting to user context and mental states.
- Wizard-of-Oz and interview studies were conducted to identify key mental states, and the system was validated in a 5-week field trial with 25 participants, showing significant improvements in smartphone usage patterns.
- This work was published on ACM CHI'24. My work included conducting the Wizard-of-Oz studies and the field trial, data analyzing, and illustrating the findings.

[P.9] [Lead] Automated Video Editing with Semantic Analysis and Aesthetic Evaluation 11/2021 - 04/2023

- I led this industry-academic collaboration project to develop an intelligent video editing system that transforms photos and videos from users' smartphones into captivating highlight reels.
- Using film editing principles, we crafted coherent narratives, emphasized key moments, and ensured seamless harmony between music and visuals.
- My work was aligning musical elements with the film's style, identifying musical climaxes and video key moments, and creating rhythm, flow, and timing for cuts and transitions.

[P.10] [Lead] Adaptive Music and Lighting Systems for Emotional Well-being 03/2022 - 03/2023

- I led this industry-academic collaboration project, developing a smart home system that dynamically adjusts music and lighting to enhance the living experience.
- Developed a framework that aligns music-emotion-light and implemented a demo using the Philips Hue system.
- My work included literature review, creating music-emotion-light framework, and realizing the demo.

[P.11] [Main Contributor] Design of Tactile Vibration Experience for Smartphones 11/2022 - 02/2023

- This industry-academic collaboration project aimed to study vibration experiences across different smartphones.
- We developed a framework that maps task urgency, importance, and metaphorical meaning to vibration timing, duration, intensity, frequency, and variability, based on a user study comparing smartphones from six brands.
- My work involved conducting literature research, developing the framework, and designing the user study.

[P.12] [Main Contributor] User's Color Preferences of Pictures Across Diverse Displays 10/2021 - 12/2022

- This industry-academic collaboration project involved six expert interviews and a user study with 89 participants to identify color preferences for various image types across different smartphone hardware.
- A framework to optimize picture color on specific hardware for improved aesthetics and user experience.
- My work included designing and conducting the user study, adjusting images, and analyzing the data.

[P.13] [Lead] Ferrofluid Speaker Design Based on Emotion-Mapped Musical Elements [Video] 09/2021 - 06/2022

- Undergraduate graduation project, where I designed and built a ferrofluid speaker that visually responds to music.
- The ferrofluid inside the speaker moves in sync with the audio, displaying a range of motions, including linear, rotational, and pulsating patterns. These movements dynamically change with the rhythm and sound of the music.

RESEARCH INTERNSHIP EXPERIENCE

Interactive Structures Lab, HCII, Carnegie Mellon University 08/2025 - Present
PhD Student / Advisor: Prof. Alexandra Ion
Work on the project of KneeExo, and lead the project of MetaBeads.

Future Lab, Tsinghua University 08/2021 - 06/2025
Research Assistant / Advisor: Prof. Yingqing Xu
Led research project of 3D-Mirrorcle [P.1], BrickSmart [P.2], Mentigo [P.3] and industry-academic collaboration project of music-lighting [P.10], and automated video editing [P.9]. Contributed to research projects [P.6] and [P.7], industry-academic collaboration project [P.11] and [P.12].

HCI Engineering Group, CSAIL, MIT 06/2024 - 11/2024
Visiting Student / Advisor: Prof. Stefanie Mueller
Worked on the project of Xstring [P.4], which focuses on 3D printing cable-driven mechanisms in a single process.

Pervasive Interaction Laboratory, Tsinghua University 10/2022 - 06/2023
Research Assistant / Advisor: Prof. Yuanchun Shi, Prof. Chun Yu
Contributed to the MindShift [P.8], using LLMs to develop interventions for healthier smartphone use.

Tencent, ID/UX Design Group 07/2021 - 10/2021
Research Intern / Advisor: Qianhui Liang
Engaged in Metaverse project, conducting market research, user analysis, and system design.

Beijing Ewaybot Technology, Robot Navigation Group 06/2020 -08/2020
Summer Intern / Advisor: Bowei Tang
Participated in Navigation algorithm research, optimizing code and conducting tests in virtual environments to improve accuracy and efficiency.

EXTRACURRICULAR ACTIVITIES

Teaching Assistant, Fundamentals of Computer Literacy 2024

Teaching Assistant, Information Art & Design Trend 2022 - 2023

President, Student Association for Science and Technology, Xinya College, Tsinghua University 2017 - 2021
Led the association to foster academic culture growth through events and innovative promotions.

Tsinghua Red Cross Society 2017 - 2018
Engaged in educational support to underprivileged rural children and organized blood donation drives.

SKILLS

CS	Python, LLM Implementation, Machine Learning, C++, C#, MATLAB, HTML, JavaScript.
EE	Embedded Systems, Arduino, Circuit Design.
Design	Fusion 360, Rhino, AutoCAD, Solidworks, Adobe Suite (proficient in PS & PR), Figma, Procreate.
Fabrication	3D Printing, Laser Cutting, CNC, Silicone Casting, Heat Sealing.

REFERENCE

Alexandra Ion (Doctoral advisor)
Assistant Professor
Director of the Interactive Structures Lab
Carnegie Mellon University
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Yingqing Xu (Master's advisor)
Professor
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